

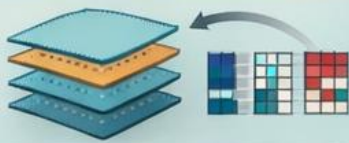
CMT 442 – Neural Networks

Module 4: Specialized Architectures and Applications

Specialized Architectures & Applications

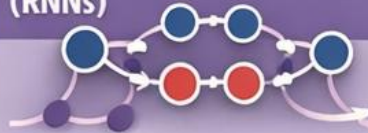
CMT 442 – Neural Networks: Module 4 Summary

Convolutional Neural Networks (CNNs)



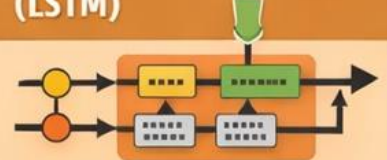
- ▶ Image Recognition
- ▶ Medical Imaging
- ▶ Quality Control

Recurrent Neural Networks (RNNs)



- ▶ Time Series
- ▶ Speech Processing
- ▶ Text Generation

Long Short-Term Memory (LSTM)



- ▶ Sequential Data
- ▶ Language Translation
- ▶ Anomaly Detection

Industry Use Cases

- ▶ Facial Recognition
- ▶ Fraud Detection
- ▶ Crop Monitoring
- ▶ Predictive Maintenance

Choosing the Right Model

- ▶ CNN for Images 
- ▶ LSTM for Sequences 
- ▶ Hybrid Approaches

Key Challenges

- ▶ Bias & Fairness 
- ▶ Data Privacy 
- ▶ Model Drift

NIST AI RMF – MAP

- ▶ Model Alignment to Risks 

ISO | ISO/IEC 42001

- ▶ Operational Controls

Responsible AI Practices

- ▶ Dataset Audits
- ▶ Explainability
- ▶ Human Oversight 

“A powerful model without governance is a liability, not an asset.”

Focus: CNNs, RNNs, LSTMs, and Industry Use Cases

Slide 1: Module Overview – Why Specialized Architectures Matter

Key Points

- Early neural networks (perceptrons, MLPs) struggle with:
 - Spatial data (images)
 - Sequential data (time, text, signals)
- Specialized architectures were designed to **match data structure**
- Module focus:
 - Convolutional Neural Networks (CNNs)
 - Recurrent Neural Networks (RNNs)
 - Long Short-Term Memory (LSTM)
 - Real-world industry applications

Teaching Emphasis

- “Architecture follows data”
- Wrong architecture = poor performance + hidden risk

Discussion Prompt

Why can't a simple feed-forward network handle images efficiently?

Slide 2: Convolutional Neural Networks (CNNs) – Core Idea

Key Concepts

- CNNs are designed for **grid-like data** (images, video frames)
- Inspired by the **human visual cortex**
- Core components:
 - Convolution layers
 - Filters (kernels)
 - Feature maps

Why CNNs Work

- Local connectivity

- Parameter sharing
- Translation invariance

Example

- Detecting edges, corners, textures before objects

Governance Note

- Feature extraction is **learned**, not explicitly defined → explainability challenge
-

Slide 3: CNN Architecture Components

Detailed Breakdown

- **Convolution Layer**
 - Applies filters to input
 - Extracts low-level → high-level features
- **Activation Function**
 - Typically ReLU
 - Introduces non-linearity
- **Pooling Layer**
 - Reduces dimensionality
 - Controls overfitting
- **Fully Connected Layer**
 - Final classification or regression

Teaching Tip

- Walk through one image → one filter → one feature map

Risk Angle

- Over-compression may remove critical features (bias risk)
-

Slide 4: CNN Applications in Industry

Key Use Cases

- Healthcare:
 - Medical imaging (X-ray, MRI, CT scans)
- Finance:
 - Document verification (ID, cheque images)

- Security:
 - Facial recognition
 - Surveillance analytics
- Manufacturing:
 - Defect detection
 - Quality control

Kenya / Africa Context

- Automated crop disease detection
- Traffic monitoring systems

Governance Link

- Image-based AI raises **privacy and consent issues**
-

Slide 5: Recurrent Neural Networks (RNNs) – Sequential Learning

Core Idea

- Designed for **sequence data**
- Output depends on:
 - Current input
 - Previous hidden state (memory)

Typical Data

- Time series
- Natural language
- Sensor data

Key Limitation

- Vanishing and exploding gradients
- Poor long-term memory

Teaching Analogy

- Reading a sentence word by word
-

Slide 6: RNN Applications and Limitations

Applications

- Speech recognition

- Text generation
- Stock price forecasting
- Predictive maintenance

Why Plain RNNs Fail

- Forget earlier information
- Unstable training for long sequences

Transition Point

- This limitation led to **LSTM networks**

Discussion Prompt

Why is remembering the beginning of a paragraph important in language tasks?

Slide 7: Long Short-Term Memory (LSTM) Networks

What Makes LSTMs Different

- Designed to solve RNN memory problems
- Uses **gated mechanisms**:
 - Forget gate
 - Input gate
 - Output gate

Key Strength

- Maintains long-term dependencies
- Stable gradient flow

Teaching Focus

- Memory cell $\neq$ hidden state
 - Gates decide *what to keep and forget*
-

Slide 8: LSTM Applications in the Real World

Use Cases

- Language translation
- Chatbots and virtual assistants
- Fraud detection (transaction sequences)
- Health monitoring (ECG, glucose patterns)

Cybersecurity Angle

- Anomaly detection in logs
- Insider threat detection

Risk & Governance

- Sequence bias
 - Training data drift over time
-

Slide 9: Architecture Selection – Matching Model to Problem

Decision Framework

Data Type	Recommended Architecture
Images	CNN
Video	CNN + RNN
Text	RNN / LSTM
Time Series	LSTM
Mixed Data	Hybrid Models

Critical Insight

- More complex ≠ better
- Model choice impacts:
 - Accuracy
 - Explainability
 - Operational risk

Standards Mapping

- **NIST AI RMF – Map**
 - **ISO/IEC 42001 – Operational controls**
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Slide 10: Responsible Deployment of Specialized Models

Key Risks

- Bias amplification (especially in vision models)
- Lack of transparency
- Data leakage

- Over-automation

Good Practice

- Model documentation
- Dataset audits
- Continuous performance monitoring
- Human-in-the-loop controls

Closing Reflection

“A powerful model without governance is a liability, not an asset.”

Module 4 Quiz: Specialized Architectures & Applications

SECTION A: QUESTIONS

Q1. Why are Convolutional Neural Networks (CNNs) better suited for image data than fully connected neural networks?

- A. They use recurrent memory to process pixels
 - B. They exploit spatial locality and shared weights
 - C. They eliminate the need for activation functions
 - D. They only work on grayscale images
-

Q2. What is the primary function of a convolutional filter in a CNN?

- A. To reduce the size of the dataset
 - B. To normalize pixel values
 - C. To extract local features from input data
 - D. To randomly drop neurons during training
-

Q3. Which CNN component mainly helps reduce dimensionality and control overfitting?

- A. Convolution layer
 - B. Activation layer
 - C. Pooling layer
 - D. Fully connected layer
-

Q4. What fundamental limitation of basic Recurrent Neural Networks (RNNs) led to the development of LSTMs?

- A. Inability to process image data
 - B. High computational cost
 - C. Vanishing and exploding gradients
 - D. Lack of activation functions
-

Q5. In an RNN, what does the hidden state represent?

- A. The final model output
 - B. The learning rate
 - C. Memory of previous inputs in the sequence
 - D. A fixed random vector
-

Q6. Which component of an LSTM decides what information should be discarded from the cell state?

- A. Input gate
 - B. Output gate
 - C. Forget gate
 - D. Activation gate
-

Q7. Which architecture is MOST appropriate for detecting anomalies in financial transaction sequences?

- A. CNN
 - B. Decision tree
 - C. LSTM
 - D. K-means clustering
-

Q8. Why is model selection considered a governance issue in AI systems?

- A. Model choice affects programming language selection
 - B. Larger models always require more data
 - C. Architecture choice influences bias, explainability, and operational risk
 - D. Only regulators are allowed to choose models
-

Q9. Which real-world application BEST demonstrates a hybrid CNN + RNN architecture?

- A. Image classification
 - B. Video activity recognition
 - C. Linear regression forecasting
 - D. Spam email filtering
-

Q10. According to responsible AI principles, which practice BEST supports trustworthy deployment of CNNs and LSTMs?

- A. Maximizing model complexity
 - B. Eliminating human oversight
 - C. Dataset auditing and performance monitoring
 - D. Using proprietary data only
-

SECTION B: ANSWERS

- A1. B** – CNNs exploit spatial locality and shared weights
- A2. C** – To extract local features from input data
- A3. C** – Pooling layer
- A4. C** – Vanishing and exploding gradients
- A5. C** – Memory of previous inputs in the sequence
- A6. C** – Forget gate
- A7. C** – LSTM
- A8. C** – Architecture choice influences bias, explainability, and operational risk
- A9. B** – Video activity recognition
- A10. C** – Dataset auditing and performance monitoring